Speed-Time Graphs

Exam code: 0625 & 0972

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Speed-time graphs

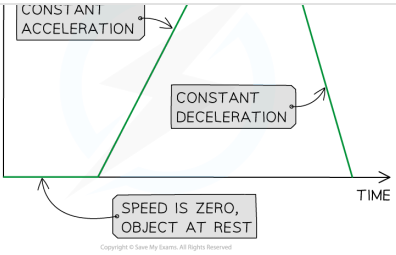
- A **speed-time graph** is used to describe the speed of an object and calculate its acceleration

Constant acceleration on a speed-time graph

- If an object is moving at a **constant acceleration**, the speed-time graph will be a **straight line**
  - If the constant acceleration is **zero**, the line will be **horizontal**
  - If the constant speed is **non-zero**, the line will have a **gradient**
- If an object has an **acceleration** of **zero**, the object is travelling at a **constant velocity**
  - Its velocity is not changing over time
  - If the **constant speed** is **zero**, then the object is **stationary**

Motion on a speed-time graph

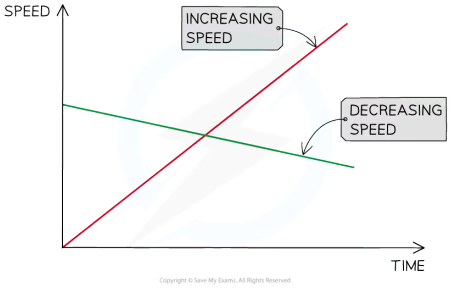
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it
Help me understand



This image shows how to interpret the slope of a speed-time graph

- The **gradient** of a speed-time graph represents the object's **acceleration**
  - A **steeper slope**, or a higher gradient, represents a **greater acceleration**
  - A **shallower slope**, or a lower gradient, represents a **slower acceleration**
- If the gradient is positive, the line slopes upward
  - A **positive gradient** represents an increasing speed, or **acceleration**
- If the gradient is negative, the line slopes downward
  - A **negative gradient** represents a decreasing speed, or **deceleration**

Speeding up and slowing down on a speed-time graph



Both of these objects are moving at a constant acceleration, because the lines are straight. The positive gradient represents an increasing speed or positive acceleration. The negative gradient represents a decreasing speed or negative acceleration.

Examiner Tips and Tricks

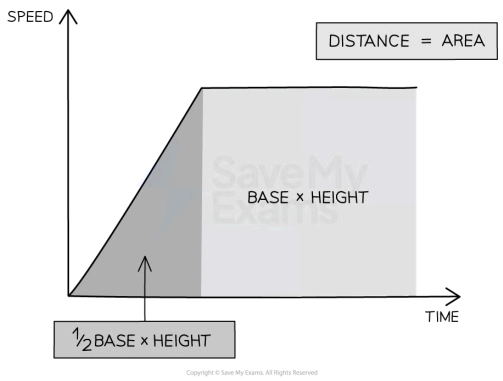
For CIE IGCSE Physics, you may be asked plot a graph of your own, or to interpret information given to you in a graph. You may also be asked to describe the motion of an object from

Using speed-time graphs

- **Speed-time graphs** can also be used to determine the distance travelled by an object

The area under a speed-time graph

- The **area** under a speed-time graph represents the **distance travelled**



The area under a speed-time graph represents the distance travelled

- If the area of a section of the speed-time graph forms a **triangle**, the area can be calculated using:

$$A_T = \frac{1}{2}bh$$

- If the area of a section of the speed-time graph forms a **rectangle**, the area can be determined using:

$$A_R = bh$$

- Where:
  - A_T = area of a triangle
  - A_R = area of a rectangle
  - b = base
  - h = height

- The **total distance travelled** can be determined by finding the **total area** under the speed-time graph
- The distance travelled for **part of the journey** can be determined by finding the area under the graph for a specific **time interval**

Area under a speed-time graph split into sections

The area under a speed-time graph can split into triangular and rectangular sections

Worked Example

The speed-time graph below shows a car journey that lasts for 160 seconds.

Calculate the total distance travelled by the car on this journey.

Answer:

Step 1: Recall that the area under a speed-time graph represents the distance travelled

- In order to calculate the total distance travelled, the total area underneath the line must be determined

Step 2: Identify each enclosed area

- In this example, there are **five** enclosed areas under the line
- These can be labelled as areas 1, 2, 3, 4 and 5, as shown in the image below:

Step 3: Calculate the area of each enclosed shape under the line

- Area 1 = area of a triangle

$$A_1 = \frac{1}{2}bh$$

$$A_1 = \frac{1}{2} \times 40 \times 17.5$$

$$A_1 = 350 \text{ m}$$

- Area 2 = area of a rectangle

$$A_2 = bh$$

$$A_2 = 30 \times 17.5$$

$$A_2 = 525 \text{ m}$$

- Area 3 = area of a triangle

$$A_3 = \frac{1}{2}bh$$

$$A_3 = \frac{1}{2} \times 20 \times 7.5$$

$$A_3 = 75 \text{ m}$$

- Area 4 = area of a rectangle

$$A_4 = bh$$

$$A_4 = 20 \times 17.5$$

$$A_4 = 350 \text{ m}$$

- Area 5 = area of a triangle

$$A_5 = \frac{1}{2}bh$$

$$A_5 = \frac{1}{2} \times 70 \times 25$$

$$A_5 = 875 \text{ m}$$

Step 4: Calculate the total distance travelled by finding the total area under the line

- Add up each of the five areas enclosed:

$$\text{total distance} = A_1 + A_2 + A_3 + A_4 + A_5$$

Examiner Tips and Tricks

Some areas will need to be split into a triangle and a rectangle to determine the area for a specific time interval, like areas 3 & 4 in the worked example above.

If you are asked to find the distance travelled for a specific time interval, then you just need to find the area of the section above that time interval.

For example, the distance travelled between 70 s and 90 s is the sum of Area 3 + Area 4

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